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A Lightweight Task-Adaptive YOLO for Tomato Ripeness Detection in Complex Orchard Environments

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Highlights

YOLOv8n–TiD detects four tomato ripeness stages in occluded and clustered orchard scenes by integrating lightweight feature extraction, dynamic upsampling, task-adaptive alignment, and F–PIoUv2 regression. The model achieves 97.9% mAP@0.5, 90.8% mAP@0.5:0.95, 1.68M parameters, and real-time Android inference above 30 FPS.

What are the main findings?

- YOLOv8n–TiD achieves high-accuracy tomato ripeness detection in complex orchard environments, reaching 97.9% mAP@0.5 and 90.8% mAP@0.5:0.95.
- The model remains lightweight, with only 1.68M parameters, 6.7 GFLOPs, and a 3.54 MB model size.

What are the implications of the main findings?

- The method effectively detects occluded, clustered, small, and visually similar tomato ripeness stages.
- The model supports real-time Android deployment above 30 FPS, providing a practical solution for robotic harvesting, field-based grading, and mobile agricultural vision systems.

Abstract

Accurate ripeness assessment of tomatoes in natural orchard settings is challenged by severe occlusion, dense clustering, and scale variation among fruits. This paper introduces YOLOv8n–TiD, a lightweight detection framework designed to overcome these obstacles. The architecture enhances the YOLOv8n backbone with a novel C2f-iRD module, which integrates re-parameterized dilated convolutions and inverted residual blocks to enlarge the receptive field while retaining fine-grained texture details. For feature fusion, we propose C2f-iRMB in the neck to ensure cross-scale consistency. To address semantic drift during upsampling, we replace standard interpolation with the DySample operator. The detection head is re-engineered as TADAH (Task-Adaptive Dynamic Alignment Head), enabling genuine task decoupling via deformable convolutions and conditionally shared features. Additionally, we introduce F–PIoUv2, a regression loss that emphasizes medium-quality predictions and curbs excessive bounding box expansion. Evaluations on a custom dataset show that YOLOv8n–TiD cuts parameters by 45%, FLOPs by 19%, and model size by 41%, while raising mAP@0.5 by 1.9 points—all in real time. On Android devices, it sustains 30 FPS inference and generalizes effectively to a distinct cherry tomato cultivar. These findings confirm the method's robustness in discriminating occluded, small, and visually similar maturity stages, providing a practical vision system for robotic harvesting and field-based grading.

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1. Introduction

As a globally significant horticultural commodity, tomato production quality heavily depends on accurate ripeness assessment during harvesting and post-harvest handling. The precision of fruit detection and maturity classification critically influences the performance of automated agricultural systems, including robotic harvesters, quality sorters, and yield prediction modules [1]. Natural growing environments—whether in open fields or greenhouses—present multiple visual challenges: intricate backgrounds, substantial occlusion from foliage, densely packed fruit clusters, subtle color variations between maturity stages, and significant size differences among targets. These complexities create substantial obstacles for developing vision algorithms that are simultaneously robust, efficient, and compact enough for field deployment [2]. The unstructured nature of tomato cultivation further complicates automated detection. High planting densities and frequent leaf concealment make it challenging to differentiate between neighboring fruits at similar ripening phases, thereby restricting the operational effectiveness of harvesting robots [3,4]. Beyond mechanized harvesting, ripeness monitoring supports broader agricultural management objectives by enabling environmental condition analysis and facilitating data-driven cultivation practices [5]. Recent advances in deep learning have demonstrated considerable potential for agricultural vision applications. Techniques spanning object detection, instance segmentation, and keypoint estimation have established viable foundations for intelligent crop management systems. Nevertheless, persistent gaps remain in achieving consistent performance across varied scenes, reliably perceiving small or overlapping fruits, and maintaining computational efficiency suitable for resource-constrained edge devices.

To address the challenges of small-object detection, occlusion, and strong illumination disturbances in natural environments, numerous studies have explored improvements within the YOLO framework. Zheng et al. [6] proposed RC-YOLOv4, an enhancement over YOLOv4 that integrates residual networks into an R-CSPDarknet53 backbone and adopts depthwise separable convolution to improve the SPP module. This design enhances the extraction of small and occluded features, achieving a detection accuracy of 94.44% under complex conditions. To balance multi-stage ripeness classification and lightweight deployment, Gao et al. [7] developed YOLOv8n-CA by incorporating Coordinate Attention, CARAFE upsampling, and a C2f-FN module, achieving high-accuracy, lightweight recognition of four-stage tomato ripeness in complex environments. Zheng et al. [8] also replaced the YOLOX backbone with DenseNet to enhance feature reuse and embedded a CBAM attention mechanism, significantly improving cherry tomato detection accuracy and efficiency for robotic systems. For natural-environment detection tasks, a lightweight model named GSBF-YOLO was proposed in [9], leveraging the GSim module, BiFPN fusion, and FocalEIoU loss to reduce computation while improving accuracy and speed in real-time field scenarios.

Liang et al. [10] introduced a YOLOv8-based lightweight model targeting cherry tomato detection. It reconstructs the backbone with LAW-Darknet53, applies adaptive downsampling to preserve small-object features, and uses an SPPFS module for enhanced multi-scale fusion, alongside a dynamic head with scale, spatial, and task-aware mechanisms to tackle occlusion, density, and lighting variations. Liu et al. [11] proposed an improved YOLOv7 model that incorporates a Swin Transformer into the backbone to capture long-range dependencies, enhances feature fusion via Trident Pyramid Networks, and applies Focaler-IoU to dynamically adjust attention to difficult samples. This design demonstrated strong robustness under diverse lighting and occlusion conditions. For greenhouse environments, Sun et al. [12] proposed the lightweight S-YOLO, which integrates a GSConv_SlimNeck to optimize the neck structure, employs an improved α-SimSPPF module for feature enhancement, and introduces a β-SIoU loss for better bounding box regression of overlapping targets, along with SE attention to focus on key features—achieving notable improvements in detection metrics for greenhouse robotic picking. Russo et al. [13] developed an autonomous robotic platform for precision viticulture by integrating mobile navigation, multimodal sensing, and AI-based leaf sampling, demonstrating the potential of combining visual perception with robotic actuation in field agricultural operations.

Fu et al. [14] tackled multi-stage tomato detection in complex environments with the YOLOv8-EA model, which uses EfficientViT as the backbone, a C2f-Faster module to accelerate fusion, SIoU for optimized box regression, and an auxiliary detection head to enhance learning capacity—achieving a balance among detection speed, accuracy, and generalization. Yong et al. [15] proposed YOLOv8-LBP based on the YOLOv8-Pose architecture, incorporating the LSKA attention module for better multi-scale feature extraction and a weighted BiFPN for fusion. This model not only detects ripe tomatoes accurately in natural scenes but also locates stems and picking points, effectively addressing occlusion and mis-segmentation issues. Finally, to reduce model complexity and dependency on GPUs, Zeng et al. [16] proposed THYOLO, a lightweight model utilizing MobileNetV3, channel pruning, and quantization. Despite significant reductions in parameters, the model maintains high accuracy and achieves 26.5 FPS on mobile platforms, providing a viable solution for low-computation deployment. Kamat et al. [17] investigated multi-class fruit ripeness detection using several object detection models, including YOLO and SSD, under natural conditions. Their study showed that deep learning-based detectors can effectively recognize different fruit ripeness stages with real-time performance, providing useful evidence for the application of object detection models in fruit quality assessment and smart harvesting.

In scenarios requiring high precision under complex occlusion and background interference, two-stage detection and image segmentation methods have demonstrated stable advantages. Wang et al. [18] developed a model based on Faster R-CNN with a ResNet-50 backbone to enhance feature representation, incorporated RoIAlign to improve bounding box localization accuracy, and integrated PANet for multi-scale feature fusion. This approach achieved outstanding performance in challenging conditions such as leaf occlusion, fruit overlap, and variable lighting, reaching a mAP of 96.14%, and provides solid technical support for precise robotic harvesting of tomatoes. Liu et al. [19] proposed a two-stage instance segmentation model called Y-HRNet by combining an improved YOLOv7 detector with HRNet. Following a detect-then-segment strategy, the model uses YOLOv7 for fast region localization and enhances HRNet with ECA attention and a DR-ASPP module to strengthen feature extraction and multi-scale fusion. This enables accurate segmentation across four ripeness stages while maintaining high efficiency, making it suitable for ripeness grading and automated harvesting applications. In addition, end-to-end detectors and hybrid Convolution–Transformer architectures have emerged as promising alternatives for complex agricultural scenarios. Wang et al. [20] introduced a lightweight detection algorithm based on an improved RT-DETR. The model adopts a PConv-based lightweight backbone, integrates deformable attention to enhance fine-detail extraction, incorporates a slimneck-SSFF structure for feature fusion, and utilizes the Inner-EIoU loss to optimize box regression. It achieves high accuracy (mAP@0.5 of 86.8%) while significantly reducing computational cost, making it suitable for real-time maturity detection in intelligent harvesting systems. Khan et al. [21] proposed a tomato ripeness recognition method and constructed the KUTomaData dataset, containing three maturity categories: green, semi-ripe, and fully ripe. Their model combines an encoder–decoder architecture with Transformer modules, achieving pixel-level ripeness segmentation under complex occlusion and lighting conditions. It significantly outperforms existing methods across multiple datasets and offers a viable technical solution for automated harvesting and sorting tasks.

Despite recent advancements, the complexity of natural environments imposes higher demands on tomato ripeness detection. Occlusion from leaves and stems, overlapping and densely clustered fruit trusses, and visual similarity across ripeness stages in terms of color and texture often lead to degraded feature representation and blurred class boundaries—resulting in missed or misclassified detections in practical applications. Although many existing methods achieve promising results under controlled conditions, they still face limitations in small-object recognition, cross-scale feature alignment, cross-variety generalization, and real-time deployment on edge devices. To address these challenges, this study focuses on detection robustness and accuracy under natural orchard conditions while also ensuring model compactness and low-latency inference—thereby providing a technical reference for deployable tomato ripeness recognition systems. We define four ripeness categories—mature, turning, breaker, and immature—and adopt YOLOv8 as the baseline framework. To enhance fine-grained texture modeling and expand the effective receptive field, we propose a novel C2f-iRD module by integrating inverted residual blocks and re-parameterized atrous convolution into the C2f backbone. For neck-level fusion, we introduce C2f-iRMB to improve cross-scale consistency and utilize DySample dynamic upsampling to suppress semantic drift caused by conventional interpolation. A task-adaptive detection head, TADAH, is constructed to achieve true decoupling of classification and regression with deformable alignment. Finally, a re-weighted loss function F–PIoUv2 is applied to enhance the learning of medium-quality and hard samples during regression optimization. Experimental results demonstrate that the proposed improvements significantly reduce parameter count and computational cost while further enhancing detection accuracy. When deployed on an Android device, the model achieves a stable frame rate exceeding 30 FPS, meeting the requirements for real-time inference on resource-constrained platforms.

The specific contributions of this study are summarized as follows:

(1) A tomato ripeness detection dataset was constructed under complex orchard conditions. The dataset covers four ripeness stages, namely mature, turning, breaker, and immature, and includes typical field challenges such as foliage occlusion, fruit overlapping, illumination variation, dense distribution, and small targets. This dataset provides experimental support for developing lightweight and deployable tomato ripeness detection models.

(2) A lightweight tomato ripeness detection framework, YOLOv8n–TiD, was developed based on YOLOv8n. Instead of simply increasing network depth or width, the proposed framework improves the accuracy–complexity trade-off by introducing task-oriented structural adaptations in the backbone, neck, upsampling stage, detection head, and bounding-box regression loss. Specifically, C2f-iRD and C2f-iRMB are embedded into the backbone and neck to enhance fine-grained feature extraction and cross-scale feature fusion while reducing computational redundancy, and DySample is introduced to improve content-aware feature reconstruction during multi-scale fusion.

(3) A task-adaptive detection head, TADAH, was designed by integrating shared convolution, task decomposition, deformable spatial alignment, and classification attention weighting. This module aims to reduce task interference between classification and regression, improve feature alignment for occluded and densely distributed tomatoes, and enhance the discrimination of visually similar ripeness stages. In addition,

F–PIoUv2 was incorporated into the regression objective to improve bounding-box localization for medium-quality, small, and partially occluded targets.

(4) It should be noted that several mechanisms used in YOLOv8n–TiD, such as iRMB, DRB, DySample, DCNv2, task decomposition, Focaler-IoU, and PIoUv2, are derived from or inspired by previous studies. Therefore, the originality of this work does not lie in independently proposing each of these mechanisms, but in their task-oriented adaptation and coordinated integration for lightweight tomato ripeness detection in complex orchard environments.

(5) Comprehensive experiments, including baseline comparison, ablation analysis, class-wise performance evaluation, confusion matrix analysis, external validation on a cherry tomato dataset, and Android edge-device deployment, were conducted to evaluate the proposed model. The results verify that YOLOv8n–TiD achieves a favorable balance among detection accuracy, model compactness, cross-variety adaptability, and real-time deployment capability.

2. Materials and Methods

2.1. Dataset Composition

The tomato image dataset used in this study was collected from the Fruit and Vegetable Planting Demonstration Garden in Jiangwang Town, Yangzhou City, Jiangsu Province. The images were captured using a Xiaomi 13 (Xiaomi Corporation, Beijing, China) smartphone. The tomato variety used was the common large red tomato. All images were taken under daylight conditions, covering diverse scenarios such as overexposure, backlighting, fruit stacking, occlusion by branches and leaves, and mixed maturity levels. The dataset includes both single-factor and multi-factor interference cases. To ensure diversity in the number of tomatoes and maturity levels within each image, the shooting distance was maintained between 30 mm and 80 mm. The image resolution was 8192 $\times$ 6144 pixels, and all images were stored in JPG format.

According to the national horticultural standard GH/T 1193-2021 [22], tomato maturity is classified into six stages based on fruit color and size: immature, mature green, breaker, early pink, pink, and fully red. In this study, we simplify the classification into four categories: immature, breaker, turning, and mature, by consolidating similar stages based on visual appearance and practical harvesting considerations. Specifically, tomatoes in the immature and mature green stages exhibit white-green to fully green surfaces and are not suitable for harvest. These stages are merged into the immature category. Tomatoes in the breaker stage show the onset of orange-red coloration with gradient transitions of green, orange, and red on the skin; this stage is defined as the breaker class suitable for long-distance transport and post-harvest ripening. Tomatoes in the early and mid-pink stages display a predominantly red appearance with some orange and minimal green; they are grouped into the turning category, appropriate for short-distance transport and near-market sale. Tomatoes in the fully red stage present uniformly red skin and are labeled as mature, ideal for immediate market distribution. The classification method for tomato maturity levels was also reviewed and validated by experts in the field of agronomy to ensure its scientific rigor and rationality. The visual characteristics and classification criteria are illustrated in Figure 1.

To improve dataset diversity and enhance the model's adaptability to complex natural environments, data augmentation was applied to the tomato image dataset. The original dataset contained 766 tomato images, all of which were manually annotated using the online annotation tool Make Sense (https://www.makesense.ai/) and exported as YOLO-format .txt label files. To ensure the accuracy and validity of the annotations, tomato targets with more than two-thirds of their area occluded by leaves, fruits, or other objects were not annotated.

Figure 1. Representative examples of tomato ripeness classification used in this study: (**a**) mature; (**b**) turning; (**c**) breaker; and (**d**) immature.

To avoid potential information leakage between the training and evaluation sets, dataset partitioning was performed before data augmentation. Specifically, the 766 original images were first divided into training, test, and validation sets at a ratio of 7:2:1. Subsequently, data augmentation was applied only to the training set, while the test and validation sets retained the original non-augmented images. The augmentation operations included random scaling and rotation, motion blur, brightness and contrast adjustment, and random noise addition. All augmentation operations were synchronously applied to the corresponding bounding-box annotations to ensure consistency and validity between augmented images and label information. The augmentation examples are shown in Figure 2. Since augmentation was applied only to the training set, augmented samples generated from the same original image did not appear simultaneously in the training, validation, or test sets, thereby ensuring the independence between model training and performance evaluation data.

Figure 2. Examples of data augmentation operations: (**a**) original image; (**b**) high brightness, motion blur, and spot occlusion; (**c**) horizontal flip, low brightness, pixel occlusion, and random noise; (**d**) symmetry transformation, high brightness, pixel occlusion, and motion blur.

After augmentation of the training set, the final dataset used for model training and evaluation contained 5728 tomato images, including 761 mature, 3752 turning, 3561 breaker, and 3389 immature tomato instances, totaling 11,463 tomato samples at different ripeness stages. The class distribution is shown in Figure 3. It should be noted that relatively colloquial class names were used in the original annotation files for practical labeling convenience. In this manuscript, all class names were standardized into the four ripeness categories defined in this study, namely mature, turning, breaker, and immature. The corresponding mapping was as follows: "fully ripened" was standardized as "mature", "ripening" as "turning", "turning" as "breaker", and "unripe" as "immature".

The number of instances among different tomato ripeness stages in the dataset is imbalanced to some extent, with the mature stage containing the fewest annotated instances. This distribution mainly originates from real plantation imaging scenarios and reflects the natural distribution of tomato ripeness stages under normal growth and practical harvesting conditions. To preserve the authenticity and field representativeness of the dataset, no forced class balancing was performed in this study; instead, the original class distribution was retained. The relatively small number of mature samples may limit, to some extent, the model's ability to learn the appearance variations in this category, especially under complex conditions such as occlusion, fruit overlapping, strong illumination, and distant small-target imaging, which may affect the stability of mature tomato detection. In contrast, immature tomatoes account for a relatively large proportion of the dataset. Since the color and texture characteristics of immature fruits are more similar to those of branches and leaves, this category is more prone to false detections and missed detections in this study. Therefore, the higher proportion of immature samples helps the model sufficiently learn the fine-grained differences between immature tomatoes and complex backgrounds, thereby improving its recognition ability for this dominant category. Overall, the class distribution not only reflects the natural ripeness status of tomatoes in real orchard scenes but also makes model training more consistent with the requirements of practical harvesting and mobile vision applications.

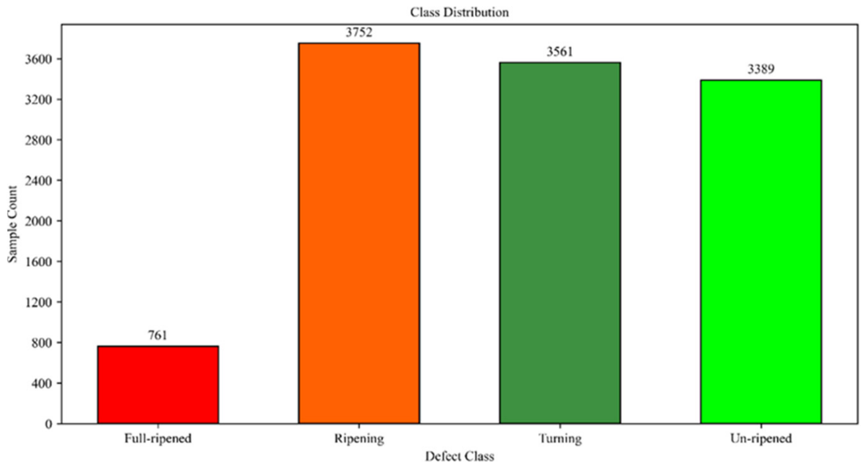

Figure 3. Class distribution of annotated tomato instances in the dataset across the four ripeness categories: mature, turning, breaker, and immature.

2.2. Baseline Model Selection

We adopted the YOLO (You Only Look Once) architecture as our detection framework based on several compelling advantages. The end-to-end design paradigm of YOLO models delivers an optimal balance between inference speed and detection accuracy, which is crucial for real-time agricultural applications. Through successive generations, YOLO variants have consistently demonstrated proficiency in addressing key challenges, including small target recognition, complex scene understanding, and multi-scale object localization. The modular design and computational efficiency of contemporary YOLO implementations further enhance their suitability for deployment on resource-constrained edge computing platforms. These attributes have propelled YOLO's widespread adoption across diverse domains such as industrial automation, precision agriculture, and intelligent transportation systems.

Recognizing that dataset-specific characteristics significantly influence model performance, we conducted an empirical evaluation of five prominent YOLO variants: YOLOv5s [23], YOLOv7 [24], YOLOv8n [25], YOLOv10n [26], and YOLOv11n [27]. All models underwent identical training protocols: 300 epochs with consistent batch configuration and hyperparameter settings. To ensure comparative fairness, we disabled the built-in Mosaic augmentation across all experiments. Training progression revealed distinct performance patterns (Figure 4). During initial phases (0–150 epochs), YOLOv5s and YOLOv7 established early accuracy advantages, while YOLOv8n, YOLOv10n, and YOLOv11n demonstrated slower convergence. Beyond epoch 200, YOLOv8n emerged as the performance leader with stable metrics, followed closely by YOLOv10n. YOLOv5s exhibited performance degradation in later stages, falling below YOLOv7, while YOLOv11n consistently trailed other variants throughout training. A comprehensive evaluation considering model complexity, computational demands, and final detection accuracy identified YOLOv8n as offering the most favorable trade-off. Its balanced architecture, combining competitive accuracy with modest resource requirements, makes it ideally suited as our baseline for subsequent architectural enhancements.

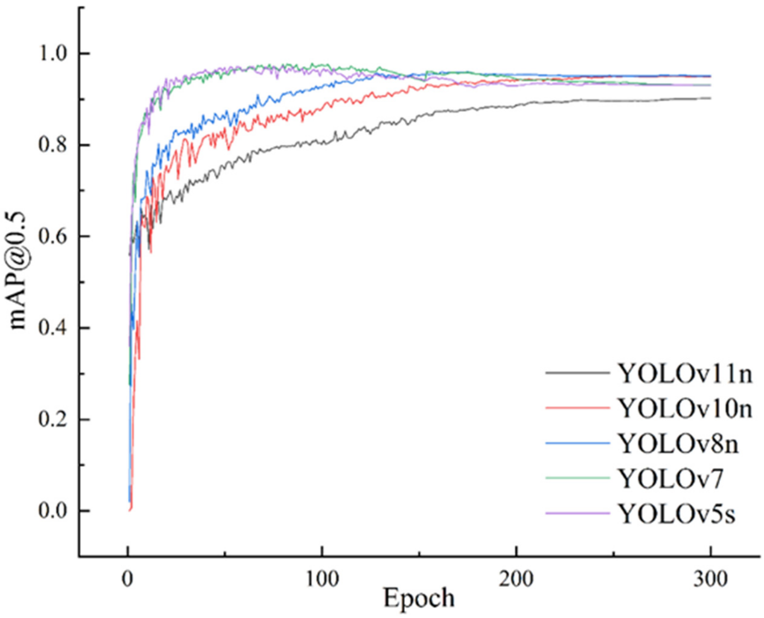

Figure 4. Training curves of mAP@0.5 for YOLOv8n and YOLOv8n–TiD on the tomato ripeness dataset.

2.3. YOLOv8n–TiD Model

2.3.1. C2f-iRD Module

To construct an efficient and accurate tomato ripeness detection model, we incorporate two advanced components into the YOLOv8 framework: the iRMB (Inverted Residual Mobile Block) [28] and the DRB (Dilated Re-parameterized Convolution Block) [29], which combine channel expansion–compression and deep convolutional representation strategies. These modules are integrated into the C2f (cross-stage partial residual) structure of YOLOv8 to form two enhanced modules: C2f-iRD, used in the backbone for feature extraction, and C2f-iRMB, employed in the neck for feature fusion. This design essentially injects multi-dilated re-parameterization capabilities into the early stages of the network, enabling broader receptive field acquisition during initial feature extraction. Meanwhile, the lightweight iRMB module supports local detail enhancement and cross-scale consistency in later layers with minimal computational overhead. By embedding C2f-iRD into the backbone, the network can capture mid- to long-range dependencies related to class boundaries at earlier stages, thus providing cleaner, more discriminative features for subsequent pyramid fusion. In the neck, following the upsampling and concatenation operations, the C2f-iRMB module effectively refines local structures and enforces semantic consistency with low cost, ensuring that the overall inference efficiency remains virtually unchanged.

Within the backbone network, the C2f-iRD module is introduced to progressively extract multi-scale features from input images, ranging from fine-grained to coarse representations. As illustrated in Figure 5, C2f-iRD centers on an internal DRB, which serves as its core unit. The DRB performs multi-level feature extraction and transformation via parallel dilated convolutions with varying dilation rates. By introducing spacing between convolutional kernel elements, these dilated operations significantly expand the receptive field without increasing the number of parameters, allowing the model to simultaneously capture fine details and global contextual patterns. The DRB also incorporates a multilayer connection mechanism that fuses features from multiple convolutional and shortcut layers, thereby enhancing the diversity and robustness of local details. This enables the network to effectively recognize subtle local variations, such as the progressive color and texture transitions of tomato surfaces from early-stage red spots to full ripeness, as well as differences in surface gloss under varying lighting conditions—all of which are critical cues for accurate ripeness classification. In parallel, the iRMB module complements the DRB by integrating a dual-path design within a unified residual framework. The local feature path employs depthwise separable convolution to efficiently model fine surface textures, while the global context path leverages window-based self-attention to capture long-range dependencies within localized regions. This design allows the model to infer occluded fruit shapes and distinguish them from complex backgrounds, such as leaves and stems. The dual-path features are then passed through a Squeeze-and-Excitation (SE) module for adaptive channel-wise recalibration, selectively emphasizing feature responses that are most relevant to maturity classification. This architecture significantly enhances the feature representation capability of the backbone, laying a solid foundation for high-quality detection in later stages.

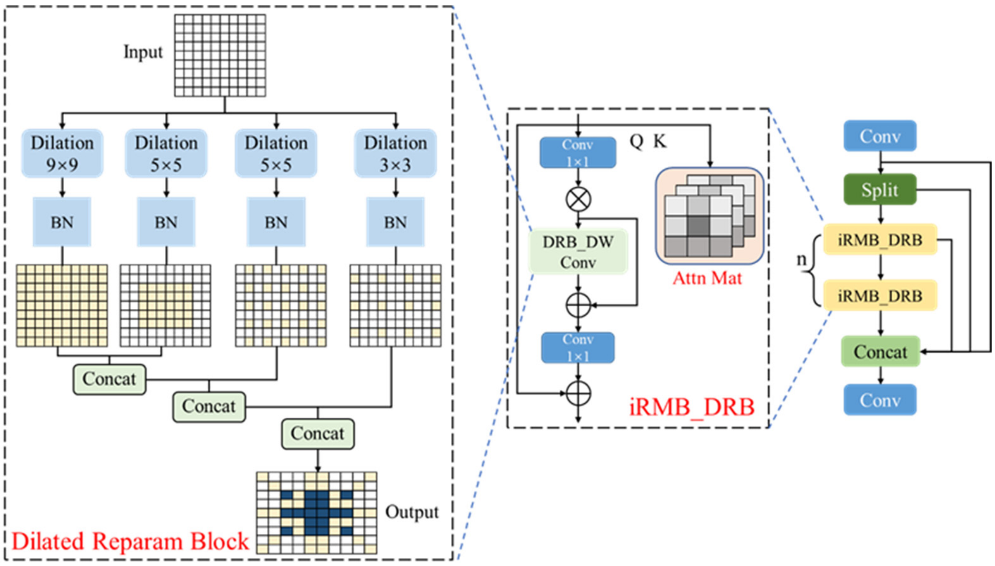

Figure 5. Architecture of the C2f-iRD module.

In the neck of the network, the C2f-iRMB module is introduced to perform deep integration and efficient fusion of multi-scale features extracted by the backbone. The original C2f structure provides rich gradient flow paths, and when combined with the iRMB's balanced local–global modeling capability, it exhibits superior performance in aligning and fusing feature maps of different resolutions. As illustrated in Figure 6, the C2f-iRMB module effectively aligns and enhances both detailed and semantic features from the backbone. It is particularly adept at addressing feature loss issues associated with small objects and partially occluded targets, such as tomatoes obscured by leaves or stems. Its window-based attention mechanism enables the establishment of long-range dependencies across scales, allowing the visible parts of occluded fruits to be associated with complementary contextual cues from other resolutions. This greatly enhances the model's robustness and accuracy in complex scenarios involving dense foliage, overlapping fruits, and varying object sizes.

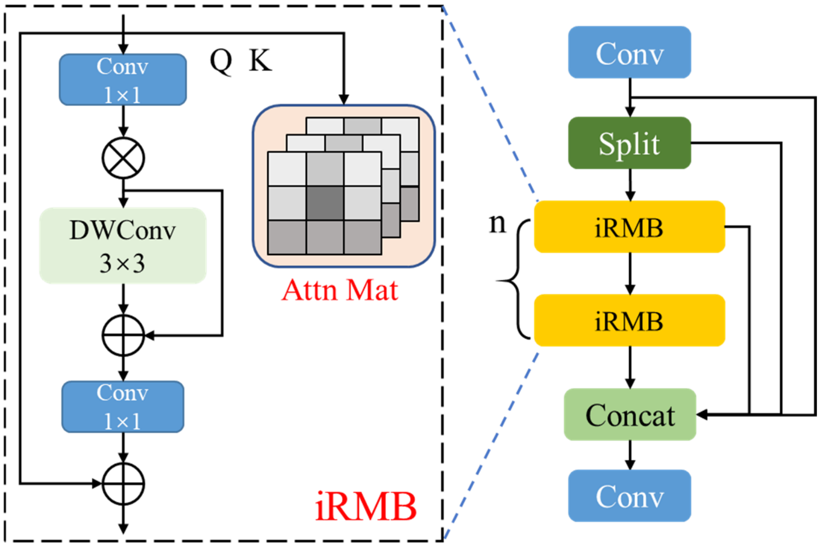

Figure 6. Architecture of the C2f-iRMB module.

2.3.2. Upsampling Operator Dysample

To improve contour accuracy and structural integrity during feature pyramid fusion, we substitute YOLOv8's default nearest-neighbor upsampling with DySample—a content-aware, dynamic point sampling operator illustrated in Figure 7. Traditional interpolation techniques depend exclusively on geometric pixel arrangements, whereas DySample [30] implements learnable, semantics-driven resampling across continuous coordinate spaces. This paradigm shift substantially enhances the upsampling module's capacity to preserve structural semantics and intricate details. Specifically, let the input low-resolution feature map be denoted as $X \in \mathrm{R}^{H\times W\times C}$. DySample first uses a lightweight predictor to generate a dynamic range factor $\delta \in [0, 0.5]$, along with an initial offset $O_{\mathrm{init}} \in \mathrm{R}^{H\times W\times 2S2}$. These are combined to yield the dynamic offset:

$$O_{dyn} = O_{int}\cdot\delta \tag{1}$$

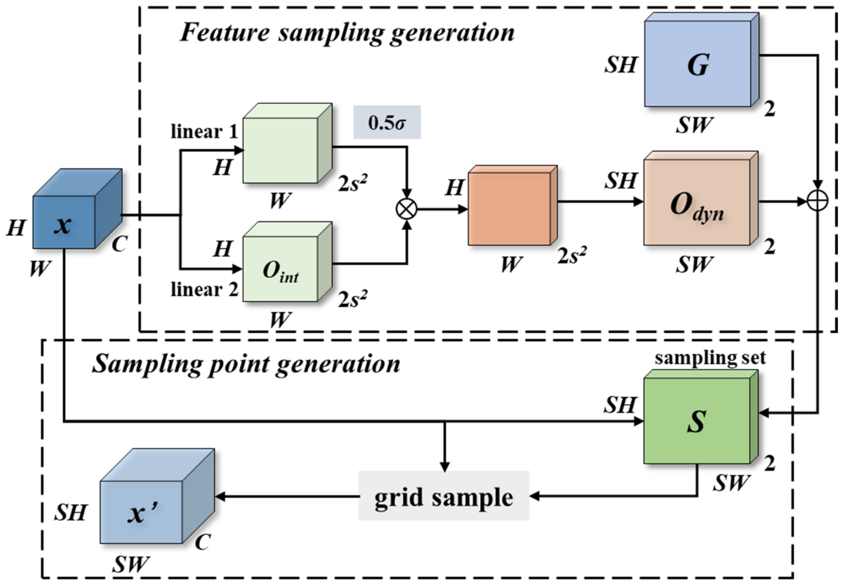

Figure 7. Network structure of the DySample upsampling module.

The resultant dynamic offset O_{dyn} modulates the predefined grid coordinates G, yielding a refined sampling lattice $S \in \mathrm{R}^{SH \times SW \times 2}$ where each entry specifies adjusted 2D sampling positions:

$$S = G + O_{dyn} \tag{2}$$

This adaptive grid configuration optimizes sampling point allocation to concentrate on semantically significant regions. The final high-resolution feature map $X' \in \mathrm{R}^{SH \times SW \times C}$ from X: is reconstructed via grid sampling:

$$X' = gridsample(X, S) \tag{3}$$

By dynamically recalibrating sampling locations, this mechanism directs computational resources toward object boundaries and high-frequency patterns. In contrast to uniform nearest-neighbor interpolation, DySample implements spatially variant processing that enhances reconstruction fidelity while maintaining semantic coherence.

This dynamic sampling mechanism adaptively adjusts the location of sampling points, allowing the upsampling process to focus more precisely on object contours and high-frequency textures. Compared to nearest-neighbor interpolation, DySample avoids applying uniform operations across all pixels, making the reconstruction more efficient and semantically aware.

The improved model benefits from content-aware sampling in high-frequency regions of tomato surfaces, such as subtle color transitions, calyx structures, and leaf veins, where DySample adjusts sampling positions and weights adaptively. This reduces oversmoothing and aliasing artifacts commonly introduced by fixed interpolation methods. During upsampling from P5 to P4 and P4 to P3 in the feature pyramid, DySample effectively suppresses aliasing and semantic drift, thereby enhancing the semantic completeness of fused multi-scale features. In challenging conditions involving occlusion by foliage, specular reflections, or uneven illumination, the dynamic sampling mechanism tends to aggregate more reliable local neighborhoods, improving the reconstruction of occluded regions and reducing confusion between immature tomatoes and background elements such as leaves or stems. This also contributes to more stable bounding box predictions by mitigating jitter caused by unreliable feature sources.

2.3.3. Task-Adaptive Dynamic Alignment Head

In YOLOv8, a decoupled head is employed to independently process classification and regression tasks. However, under complex conditions—such as color-similar targets, densely clustered objects, occlusion from foliage, and strong specular reflections—this architecture often shows performance limitations. First, the shared features are not conditionally recalibrated for the specific needs of classification and regression. As a result, classification branches may over-focus on geometric cues, while regression branches may be distracted by textural patterns, leading to task entanglement and maturity-level confusion. Second, after upsampling and cross-layer concatenation, the spatial misalignment between feature maps and the blurring of object boundaries become evident. Standard convolutions lack the adaptability to precisely align with true object contours, causing issues such as duplicate bounding boxes, box jitter, and limited improvements in average precision under mid-to-high IoU thresholds. Moreover, in cluttered backgrounds, the textured leaf regions and high-light surfaces are frequently misactivated by the classification branch, deteriorating overall detection precision and affecting the smoothness of precision–recall and confidence–accuracy curves.

To address the performance bottlenecks caused by multi-task entanglement and geometric misalignment in tomato ripeness detection, we propose a novel detection head termed the TADAH (Task-Adaptive Dynamic Alignment Head). The TADAH first applies a shared convolution to the multi-scale feature maps from P3 to P5, enabling cross-scale feature interaction and fusion. This design not only simplifies the feature processing path and reduces computational cost and parameter count but also maintains or even improves detection performance, making it more suitable for deployment on mobile or low-resource platforms. Subsequently, a task decomposition mechanism is introduced to split the fused feature representation into two dedicated branches—one for classification and one for regression. This allows the classification branch to focus on visual cues such as color and texture, while the regression branch concentrates on geometric and boundary information. This separation mitigates cross-task interference and leads to improved classification accuracy and bounding box precision [31,32].

In the regression branch, we introduce DCNv2 (Dynamic Convolution v2) [33] to better fit the rounded contours of tomato fruits and accommodate deformation caused by leaf occlusion. The required offsets and modulation masks are adaptively generated from the fused features, enabling the convolutional sampling positions and weights to dynamically adjust based on image structure. This allows for robust boundary alignment and effectively reduces bounding box drift and jitter during inference. In the classification branch, we leverage the shared interactive features to perform dynamic feature selection and align classification confidence with localization quality. On one hand, adaptive weighting enhances sensitivity to maturity-related visual cues such as color distribution, surface gloss, and micro-textures. On the other hand, this design ensures that high-confidence predictions are more likely to correspond to high-quality bounding boxes, thereby improving the reliability of ripeness classification. To accommodate variations in input resolution and fruit size, a scaling layer is added at the end of the detection head. This layer dynamically calibrates numerical outputs and response magnitudes across different scales, enhancing the stability and generalization of the model under multi-scale input conditions.

As shown in Figure 8, the feature maps from P3, P4, and P5 are sequentially fed into two consecutive 3×3 shared convolution layers to perform cross-scale fusion. The output of the first convolution is concatenated with a shortcut from the second layer, and the resulting fused feature is split into four branches. Two of these branches implement task decomposition [34], where global average pooling, fully connected layers, and Sigmoid activation are used to generate channel-wise gating signals, producing task-specific features for classification and regression, respectively. Another branch generates the modulation mask and sampling offset required by DCNv2, which are then combined with the regression-specific features and passed into a deformable convolution to perform geometric alignment. The final branch generates a weight map for the classification task, which is element-wise multiplied with the classification-specific features to suppress false activations caused by leaf specularities, shadows, and other irrelevant signals, thereby enhancing foreground consistency. The detection head outputs predictions at three scales, and a scaling layer is applied at the end to adaptively calibrate the output magnitudes across different stride levels. During this process, the offset dynamically adjusts the sampling positions of the convolution kernels to better fit local structures, while the mask modulates the contribution of each sampling point. Together, these mechanisms enable dual-level alignment in both spatial geometry and attention weighting, allowing the network to better capture key visual cues and boundary features relevant to tomato ripeness.

Compared with the original detection head, the proposed design effectively mitigates the adverse impact of task coupling and spatial misalignment on both training and inference, without significantly increasing parameter count or computational overhead. In complex tomato ripeness detection scenarios—characterized by similar color tones, densely clustered instances, and occlusions—the improved head achieves enhanced recall and average precision for small objects, reduces misclassification between adjacent ripeness levels, smooths the high-recall segment of the precision–recall (PR) curve, and yields more balanced mAP across varying IoU thresholds. Moreover, the additional computational cost mainly arises from a single deformable convolution in the regression branch and a few small-kernel convolutions. By leveraging shared convolutional pathways and 1×1 prediction heads, the model remains operator-friendly and maintains real-time performance, aligning well with the design goal of lightweight deployment.

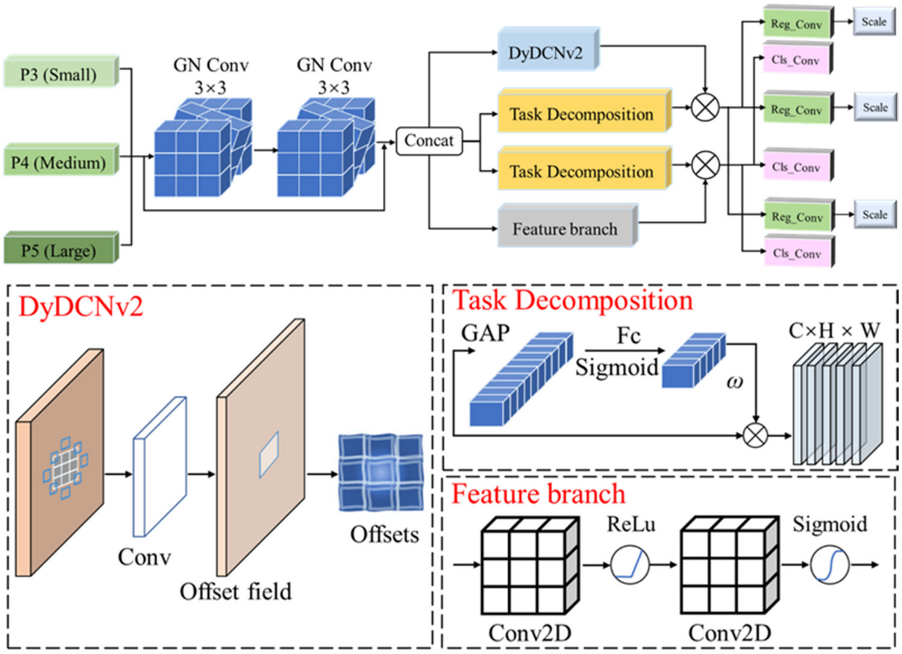

Figure 8. Task-adaptive dynamic alignment head structure.

To make the implementation of TADAH more explicit, Algorithm 1 summarizes its forward propagation process. For each detection scale, the input feature is first processed by shared convolutional layers to obtain a stacked shared feature representation. This feature is then decomposed into classification-specific and regression-specific branches, followed by task-adaptive alignment for the two prediction tasks. The regression branch uses deformable offsets and modulation masks for spatial alignment, whereas the classification branch uses a spatial attention weight map to emphasize maturity-related discriminative regions.

In TADAH, offsets, modulation masks, and attention weights are generated from the stacked shared feature F_i. For regression alignment, a 3×3 convolution produces $3k^2$ output channels, where $k = 3$. The first $2k^2$ channels are used as deformable offsets, and the remaining k^2 channels are transformed by a Sigmoid function to obtain modulation masks. These offsets and masks are applied to the regression-specific feature through DyDCNv2, allowing the regression branch to better adapt to tomato boundary variations caused by occlusion, overlap, and shape changes. For classification alignment, a lightweight 1×1 convolution, ReLU activation, 3×3 convolution, and Sigmoid activation are used to generate a one-channel spatial attention map. This map is multiplied by the classification-specific feature to enhance maturity-related fruit responses and suppress irrelevant background activations. In addition, the task decomposition module generates layer attention weights through global average pooling and two 1×1 convolutions, enabling the classification and regression branches to adaptively select different shared features according to their task requirements.

Algorithm 1. Task-adaptive alignment process of the TADAH module

Input: Stacked shared feature $F_i \in R\hat{}(B \times C \times H_i \times W_i)$ at the i-th detection scale; kernel size k = 3.
Output: Spatially aligned regression feature $F_i$reg and attention-refined classification feature $F_i$cls.

Compute layer attention weights for task decomposition

```
Compute global descriptor
G_i ← GAP(F_i)
Generate layer attention weights for task decomposition
A_i cls ← Sigmoid(Conv1×1cls,2(ReLU(Conv1×1cls,1(G_i))))
A_i reg ← Sigmoid(Conv1×1reg,2(ReLU(Conv1×1reg,1(G_i))))
Generate task-specific features
F_i cls ← TaskDeccls(F_i, A_i cls)
F_i reg ← TaskDecreg(F_i, A_i reg)
Predict offsets and modulation masks for regression alignment
Z_i ← Conv3×3om(F_i)
(ΔP_i, M_i raw) ← Split(Z_i; 2k^2, k^2)
M_i ← Sigmoid(M_i raw)
Perform deformable spatial alignment for regression
F_i reg ← DyDCNv2(F_i reg, ΔP_i, M_i)
Generate spatial attention weights for classification alignment
A_i spa ← Sigmoid(Conv3×3(ReLU(Conv1×1(F_i))))
Refine classification-specific features
F_i cls ← F_i cls ⊙ A_i spa
Return aligned features
return F_i reg, F_i cls
```

2.3.4. F–PIoUv2 Loss Function

Tomatoes are often cultivated in complex environments where fruits are frequently occluded by leaves, branches, or other fruits. Such conditions pose significant challenges for accurate ripeness recognition. The YOLOv8n model employs the CIoU (Complete IoU) loss as its bounding box regression function; however, certain penalty terms within CIoU can cause the anchor boxes to expand excessively during the regression process. This behavior negatively impacts convergence speed and may lead to suboptimal localization performance, especially in densely occluded scenarios.

To enhance the convergence speed and accuracy of tomato ripeness detection, this study proposes a novel loss function termed F–PIoUv2, which integrates the core ideas of Focaler IoU [35] and PIoUv2 [36]. Specifically, PIoUv2 effectively mitigates the issue of anchor box enlargement toward the target location during the regression process, which often slows down model convergence. Meanwhile, Focaler IoU alleviates the negative impacts of sample imbalance and varying sample difficulty on bounding box regression. By incorporating an adaptive penalty factor and interval mapping mechanism, the proposed F–PIoUv2 loss function accelerates convergence and enhances the robustness and accuracy of tomato ripeness detection under complex environmental conditions.

Focaler IoU constructs the IoU (Intersection over Union) using a piecewise function to improve the performance of bounding box regression. The corresponding formulation is defined as follows:

$$L_{\mathrm{IoU}^{\mathrm{focaler}}} = \begin{cases} 0, & L_{\mathrm{IoU}} < d \\ \frac{L_{\mathrm{IoU}} - d}{u - d}, & d \leq L_{\mathrm{IoU}} \leq u \\ 1, & L_{\mathrm{IoU}} > u \end{cases} \quad (4)$$

In the formula, L_{IoU} represents the initial value, while d and u are constrained within the range [0, 1]. Different regression samples correspond to different values of d and u. The detailed formulation is as follows:

$$X_{\mathrm{IoU}^{\mathrm{focaler}}} = 1 - L_{\mathrm{IoU}^{\mathrm{focaler}}} \quad (5)$$

By analyzing the formula of Focaler IoU, it can be observed that when the initial IoU value (L_{IoU}) falls below a certain threshold, the corresponding $L_{\mathrm{IoUfocaler}}$ becomes zero. However, in this study's dataset, many tomato targets are small and of low quality due to complex backgrounds. Such behavior is clearly inconsistent with the actual requirements of this task. Therefore, we adopt an improved formulation that still follows a piecewise function structure but eliminates the two original parameters. This allows for more comprehensive utilization of sample data, ensuring that even low-quality tomato instances are properly accounted for. The improved formulation enables a more thorough analysis of object locations and facilitates more stable model convergence. It is thus better suited for the tomato ripeness detection task in this dataset. The proposed loss function is defined as in Equations (6) and (7).

$$L_{\mathrm{IoU}^{\mathrm{f}}} = \begin{cases} \frac{L_{\mathrm{IoU}}}{u}, & L_{\mathrm{IoU}} \leq u \\ 1, & L_{\mathrm{IoU}} > u \end{cases} \quad (6)$$

$$X_{\mathrm{IoU}^{\mathrm{f}}} = 1 - L_{\mathrm{IoU}^{\mathrm{f}}} \quad (7)$$

Most commonly used bounding box regression loss functions guide the anchor box to gradually expand until it fully covers the ground truth and then contract to achieve linear regression. This process often leads to increased training time and may compromise localization accuracy. The PIoU (Powerful-IoU) loss function effectively addresses this issue by avoiding unnecessary expansion of the anchor box. Specifically, it adaptively selects penalty factors based on the target size and adjusts the gradient according to the quality of the predicted box. By directly minimizing the distance between the four edges of the anchor box and the ground truth box, PIoU guides the regression process along a nearly straight path toward the target. As a result, it enables faster and more stable convergence compared to other IoU-based losses. In this study, we introduce an improved version named PIoUv2, which retains the core advantages of PIoU while further optimizing the regression trajectory. By incorporating a target-size-aware penalty term and a gradient adjustment function conditioned on the box quality, PIoUv2 facilitates effective and efficient bounding box regression, achieving faster convergence and higher localization precision than existing IoU-based loss functions. The mathematical formulation of the PIoU loss is presented as follows:

$$P = \left(\frac{dw_1}{w_{gt}} + \frac{dw_2}{w_{gt}} + \frac{dh_1}{h_{gt}} + \frac{dh_2}{h_{gt}}\right)/4 \quad (8)$$

$$X_{PIoU} = 1 - PIoU = L_{IoU} + 1 - e^{-P^2} \quad (9)$$

In the equation, P denotes a penalty factor that adapts to the target size, reflecting the function's adaptability to object scale. When used as a penalty term in the loss function, it effectively avoids the undesirable enlargement of the predicted bounding boxes. Specifically, dw_1, dw_2, dh_1, and dh_2 represent the absolute distances between the corresponding edges of the predicted and ground truth boxes, while w_{gt} and h_{gt} denote the width and height of the ground truth box, respectively. The function $f(x)$ is designed to adaptively adjust the gradient magnitude based on the quality of the predicted bounding box.

To further enhance the focus on medium-quality detection boxes, a non-monotonic attention function is incorporated into the PIoU loss, inspired by the focal mechanism. This leads to the formulation of a novel loss function, termed PIoUv2, which improves the model's sensitivity to moderate-quality predictions while maintaining robustness to high- and low-quality samples. The specific formulation of the PIoUv2 loss function is as follows:

$$q = e^{-P},\ \mathrm{q} \in (0, 1] \tag{10}$$

$$m(x) = 3x \cdot e^{-x^2} \tag{11}$$

$$X_{PIoUv2} = 3m(\lambda q) \cdot X_{PIoU} \tag{12}$$

In the PIoUv2 formulation, $m(\lambda q)$ denotes a non-monotonic attention function, where q substitutes the original penalty term P and reflects the predicted box quality. The value of q lies within the interval (0, 1], and λ is a hyperparameter that controls the behavior of the attention function, typically selected from the range [1.1, 1.7]. The optimal value of λ should be re-evaluated depending on the characteristics of the target dataset. By incorporating this non-monotonic attention function, PIoUv2 enhances the model's focus on medium-quality detection boxes, improving localization precision in complex environments. Importantly, PIoUv2 introduces only a single additional hyperparameter λ, which simplifies the model tuning process.

To improve the clarity of the loss formulation, the notation of F–PIoUv2 is unified in this section. Let B and Bgt denote the predicted bounding box and the corresponding ground-truth box, respectively. The intersection-over-union between them is defined as s = IoU(B, Bgt), where $s \in [0, 1]$. The PIoUv2 loss introduces a target-size-aware penalty term P (B, Bgt), which is calculated according to the normalized distances between the four corresponding edges of the predicted and ground-truth boxes. This penalty term constrains the regression trajectory and reduces unnecessary enlargement of the predicted box during optimization. A box-quality factor $q \in (0, 1]$ is further derived from the penalty term to describe the localization quality of the predicted box. Based on this quality factor, the non-monotonic attention function m(λq) is used to adjust the contribution of samples with different localization qualities, where λ is a hyperparameter controlling the attention response.

Accordingly, the F–PIoUv2 loss is formulated as

$$L_{F-PIoUv2} = \omega(s,\ q;\ \lambda) \cdot L_{PIoUv2} \tag{13}$$

where $\mathrm{L}_{\mathrm{PIoUv2}}$ denotes the original PIoUv2 regression loss, and ω (s, q; λ) represents the focal weighting term determined by the IoU value s, the box-quality factor q, and the attention-control parameter λ. Through this weighting strategy, F–PIoUv2 increases the regression emphasis on samples that still contain useful localization information but are not yet well aligned with the ground truth while reducing the excessive dominance of very low-quality or already well-localized samples.

The overall training loss is defined as

$$L_{total} = \alpha \mathrm{L}_{box} + \beta \mathrm{L}_{cls} + \gamma \mathrm{L}_{\mathrm{DFL}} \tag{14}$$

where L_{box} = $\mathrm{L}_{F\text{-}PIoUv2}$, L_{cls} denotes the classification loss, and $\mathrm{L}_{\mathrm{DFL}}$ denotes the distribution focal loss for bounding-box distribution learning. The coefficients α, β, and γ are the loss weights used to balance the three terms. Thus, the model minimizes the weighted sum of localization, classification, and distribution regression objectives during training, rather than optimizing F–PIoUv2 independently.

To further explain the optimization behavior of F–PIoUv2, the theoretical weight response and its corresponding gradient were analyzed under different values of λ, as shown in Figure 9. Figure 9a shows that the weight response first increases and then decreases as the box-quality factor q increases, indicating a clear non-monotonic weighting behavior. This suggests that F–PIoUv2 does not continuously increase the optimization weight for samples with higher localization quality. Instead, it assigns relatively higher weights to medium-quality bounding boxes, which usually contain reliable localization information but still require further boundary refinement. For extremely low-quality predictions, the weight remains relatively small, reducing the influence of unreliable regression samples. For high-quality predictions, the weight gradually decreases, preventing already well-aligned boxes from dominating the optimization process.

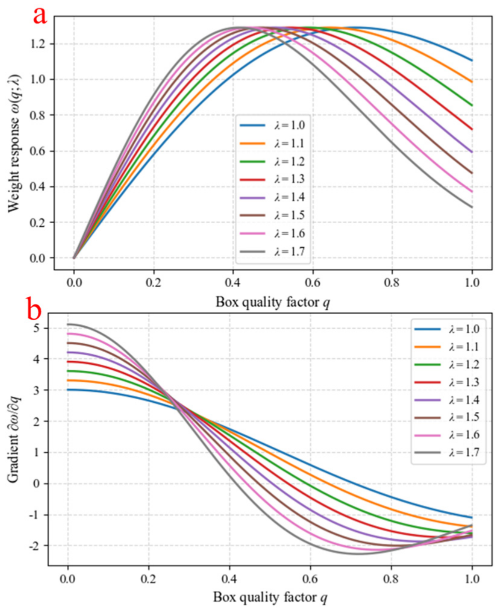

Figure 9. Theoretical weighting behavior of F–PIoUv2 under different values of λ: (**a**) weight response curve; (**b**) gradient behavior of the weight term with respect to the box-quality factor q.

The influence of λ can also be observed from the weight response curves. As λ increases from 1.0 to 1.7, the peak of the curve shifts toward a lower q region, and the descending part of the curve becomes steeper. This indicates that λ controls the quality interval emphasized by the loss function. A larger λ makes the loss focus more on relatively lower-quality but still informative samples, whereas a smaller λ shifts the emphasis toward higher-quality samples. Therefore, λ provides a flexible mechanism for adjusting the regression attention range according to the difficulty of the detection task.

Figure 9b further shows the gradient behavior of the weight term with respect to q. The gradient is positive in the low-quality interval, indicating that the weight increases as the localization quality improves. After the curve reaches its peak, the gradient becomes negative, meaning that the weight is gradually reduced for high-quality samples. This positive-to-negative transition is consistent with the non-monotonic response observed in Figure 9a. Such a gradient pattern helps F–PIoUv2 allocate more effective regression emphasis to medium-quality samples while suppressing excessive contributions from poorly matched or already well-localized boxes. This property is beneficial for tomato ripeness detection under complex orchard conditions, where small targets, fruit overlap, partial occlusion, and dense distribution often lead to unstable boundary regression.

In summary, the improved detection model—YOLOv8n–TiD—is constructed based on the integration of multiple lightweight and adaptive components. The overall network architecture is illustrated in Figure 10.

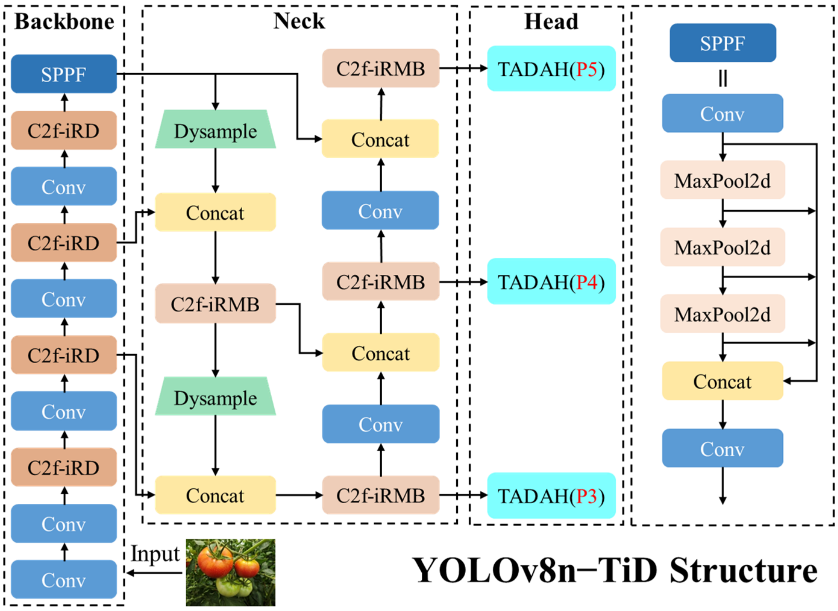

Figure 10. YOLOv8n–TiD network structure.

3. Results

3.1. Experimental Parameters and Evaluation Indicators

The experimental platform configuration is detailed in Table 1. We established a standardized training protocol to ensure reproducible and comparable results across all model evaluations. As shown in Table 2, Input images were uniformly resized to 640 × 640 pixels, and models underwent 300 training epochs with a batch size of 16. The optimization configuration employed an initial learning rate of 0.001 with cosine annealing scheduling, maintaining the same initial rate. For detection validation, we utilized an IoU (Intersection-over-Union) threshold of 0.5 to determine positive matches. To eliminate potential confounding factors, we disabled the built-in Mosaic augmentation functionality across all YOLO variants while preserving other default parameter settings, thereby ensuring experimental consistency.

Table 1. Experimental platform parameters.

Name	Details
CPU	i5-14600KF
GPU	NVIDIA RTX 4080
RAM	32 GB
VRAM	16 GB
Operating system	Windows 10
Programming language	Python 3.9
Deep Learning Library	PyTorch 2.1.0

Model assessment followed a dual-criteria framework examining both detection precision and computational efficiency. Detection accuracy was quantified through mAP (mean Average Precision), while computational characteristics were evaluated using parameter count, GFLOPs (floating-point operations), and model storage footprint. This comprehensive metric set enables thorough analysis of each architecture's detection capability alongside its practical deployment feasibility, particularly for resource-constrained agricultural applications.

Table 2. Experimental training parameters.

Name	Details
Epochs	300
Batch size	16
Initial learning rate	0.001
Final learning rate factor	0.01
Momentum	0.937
Weight decay	0.0005
optimizer	SGD
Mosaic	OFF

3.2. Experimental Results

3.2.1. Comparison Experiment of Upsampling Module

To evaluate the effectiveness of the DySample upsampling operator on our tomato-maturity dataset, we conducted a controlled comparison in which the default nearest-neighbor upsampling in the YOLOv8n baseline was replaced with either DySample or the widely adopted CARAFE [37]. The results are reported in Table 3. Compared with the baseline, the DySample-equipped model exhibits stronger lightweight characteristics: it reduces both parameter count and FLOPs without materially increasing the model file size, while also yielding a higher mAP, thereby improving detection accuracy. In contrast, substituting CARAFE leads to substantial increases in parameters and computation, a markedly larger model size, and a slight decline in mAP, indicating that the added architectural complexity does not translate into performance gains on this dataset. Collectively, these findings show that DySample achieves a more favorable efficiency–accuracy trade-off and is thus better suited to tomato-maturity detection under resource-constrained deployment scenarios.

Table 3. Comparison of upsampling performance.

Models	Parameter/M	GFLOPs	Model Size/MB	$mAP_{@0.5}$ (%)
YOLOv8n	3.05	8.2	5.99	96.0
YOLOv8n-CARAFE	3.20	8.8	6.23	95.9
YOLOv8n-DySample	3.01	8.1	5.98	96.9

3.2.2. Comparison Experiment of Detection Head

To verify the effectiveness of the TADAH on the proposed tomato maturity dataset, we conducted a comparative experiment by replacing the detection head in the YOLOv8n baseline with the Dyhead module [38]. The experimental results are presented in Table 4. It can be observed that although introducing TADAH leads to a slight increase in computational cost, the number of parameters and model weight size are significantly reduced, demonstrating strong lightweight characteristics. Moreover, the model achieves a slightly higher mAP compared to the baseline. In contrast, the model equipped with Dyhead, which integrates multiple attention mechanisms to enhance feature representation, incurs a significantly increased model complexity yet fails to improve detection accuracy. This suggests that Dyhead does not exhibit good adaptability on this specific dataset. Compared to the original decoupled head, TADAH effectively mitigates the impact of task coupling and spatial misalignment during training and inference without substantially increasing model complexity. In scenarios characterized by similar color, high-density clustering, and heavy occlusion among tomato fruits, TADAH demonstrates improved target recall and average precision, reduced confusion between adjacent maturity levels, and smoother precision–recall curves in the high-recall region. It also maintains balanced AP under different IoU thresholds. The additional computational overhead mainly stems from a single deformable convolution in the regression branch and a few lightweight convolutional layers, while the use of shared pathways and 1×1 convolution prediction heads ensures operator-friendly deployment and real-time inference, satisfying the requirements of lightweight model design. In summary, the proposed TADAH detection head demonstrates outstanding efficiency and detection performance, making it better suited for the tomato maturity dataset used in this study.

Table 4. Comparison of detection heads' performance.

Models	Parameter/M	GFLOPs	Model Size/MB	$\mathbf{mAP_{@0.5}}$ (%)
YOLOv8n	3.05	8.2	5.99	96.0
YOLOv8n-Dyhead	3.63	10.5	6.90	96.0
YOLOv8n-TADAH	2.25	8.7	4.48	96.4

3.2.3. Analysis of F–PIoUv2 Hyperparameter Values

In this study, the F–PIoUv2 loss function is employed, which incorporates the design philosophy of PIoUv2 and therefore retains a hyperparameter λ to control the behavior of the attention function. The value of λ is set within the range of 1.1 to 1.7. To determine the optimal value of this hyperparameter, we conducted a series of ablation experiments on the proposed model while keeping all other improvements unchanged. Specifically, the model was trained using different λ values, and its performance was evaluated on the tomato maturity dataset. The experimental results are summarized in Table 5. It is evident from the results that when λ = 1.5, the model achieves the most significant performance improvement, indicating that this setting is the most suitable for the characteristics of the current dataset.

Table 5. The comparative analysis under different λ values.

No.	Models	$\mathbf{mAP_{@0.5}}$ (%)
1	$\lambda = 1.1$	96.8
2	$\lambda = 1.2$	97.3
3	$\lambda = 1.3$	96.3
4	$\lambda = 1.4$	97.4
5	$\lambda = 1.5$	97.8
6	$\lambda = 1.6$	97.3
7	$\lambda = 1.7$	97.1

In addition to evaluating the effect of different λ values, this study also compared the performance of several commonly used loss functions on the improved model. The comparison included DIoU, SIoU, EIoU, PIoU, and PIoUv2, as summarized in Table 6. According to the detection accuracy curves of each loss function, PIoUv2 exhibited the best performance among them. Based on this observation, we incorporated the idea of Focaler IoU into PIoUv2 and proposed a novel hybrid loss function, F–PIoUv2. As illustrated in Figure 11, the value of mAP@0.5 gradually converged after 200 epochs of training, and the proposed loss function ultimately achieved the highest mAP score. This result confirms that a well-designed loss function can provide substantial performance gains to detection models. In summary, the F–PIoUv2 loss function integrates the advantages of both PIoUv2 and Focaler IoU. By combining the adaptive penalty mechanism of PIoUv2 and the piecewise linear mapping strategy of Focaler IoU, F–PIoUv2 accelerates model convergence and significantly improves the performance of tomato ripeness detection under complex conditions.

Table 6. Comparison of different loss functions.

Name	$mAP_{@0.5}$ (%)
DIoU	97.4
SIoU	97.0
EIoU	97.5
PIoU	96.7
PIoUv2	97.6
F–PIoUv2	97.8

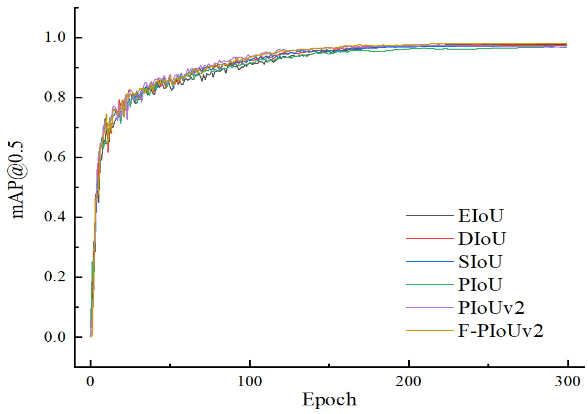

Figure 11. Detection performance comparison of YOLOv8n–TiD using different bounding-box regression loss functions.

3.2.4. Ablation Experiment

To evaluate the individual contributions of each proposed component—including the TADAH detection head, DySample upsampling operator, C2f-iRD and C2f-iRMB feature extraction and fusion modules, and the F–PIoUv2 loss function—an ablation study was conducted. The experimental results are summarized in Table 7. In this study, the ablation study was designed from the perspective of the accuracy–complexity trade-off rather than accuracy improvement alone. The progressive cumulative ablation strategy was adopted not only to evaluate the performance variation after introducing each module, but more importantly, to analyze the synergistic contribution of different modules within the overall lightweight framework. The experimental results show that, with the gradual incorporation of the proposed components, the model does not simply become more complex due to structural modifications. Instead, the number of parameters, computational cost, and model size generally decrease, while detection accuracy progressively improves. This indicates that the proposed architectural design does not rely on increasing model scale to obtain higher accuracy. Rather, it achieves simultaneous model simplification and performance enhancement by reducing redundant computation and optimizing the feature representation pathway. These results are also consistent with the objective of this study for mobile and edge-device deployment, namely improving the accuracy and stability of tomato ripeness detection in complex natural environments while maintaining real-time inference capability. The results show that integrating DySample, TADAH, C2f-iRD, and C2f-iRMB individually led to varying degrees of model simplification. For instance, although introducing the TADAH detection head slightly increased the computational cost, this was due to the integration of DyDCNv2, which learns both offsets and modulation masks to dynamically adjust convolution sampling locations and weights. This adaptive mechanism enhances spatial alignment with the tomato contours, especially under occlusions. Replacing the original C2f blocks with C2f-iRD in the backbone and C2f-iRMB in the neck further improved both efficiency and precision. This is mainly attributed to the use of factorized depthwise separable convolutions combined with pointwise projections and the DRB strategy, which reduces kernel complexity and branch redundancy. Additionally, windowed attention and multi-dilation convolution operations improve boundary localization, enabling higher-quality multi-scale features under constrained resources. Building upon this, the introduction of TADAH offered additional benefits. By replacing decoupled thick towers with shared convolution and task decomposition and applying DyDCNv2 in the regression branch and probability modulation in the classification branch, the detection head effectively disentangled spatial and task-level dependencies. This led to enhanced accuracy while significantly reducing parameter count and computational overhead. The DySample upsampling operator further contributed to the lightweight design. By generating learnable offsets through 1 × 1 lightweight convolutions, DySample enables content-adaptive sampling while preserving edge textures and spatial details during upsampling, improving the alignment of multi-scale features. Finally, the integration of the F–PIoUv2 loss function—a hybrid design combining Focaler IoU and PIoUv2 with a tuned hyperparameter $\lambda = 1.5$—provided refined gradient focus and tight localization constraints, improving convergence stability and enhancing both classification and localization performance. Overall, the improved YOLOv8n–TiD model achieved a 45% reduction in parameter count, a 19% decrease in FLOPs, and a 41% reduction in model size, while improving mAP by 1.9 percentage points compared to the original YOLOv8n baseline. The final model shows robust performance and a compact design, making it highly suitable for tomato ripeness detection in complex agricultural environments.

Table 7. Ablation test results of the YOLOv8n–TiD model.

Models	Parameter/M	GFLOPs	Model Size/MB	$mAP_{@0.5}$ (%)	mAP@0.5:0.95 (%)
YOLOv8n	3.01	8.2	5.99	96.0	88.7
+DySample	3.01	8.1	5.98	96.9	90.0
+TADAH	2.25	8.7	4.48	96.4	90.1
+C2f-iRMB	2.77	7.7	5.29	96.7	88.6
+C2f-iRD	2.80	8.0	4.57	96.2	88.4
+C2f-iRD-C2f-iRMB	2.30	6.6	4.76	96.6	88.8
+C2f-iRD-C2f-iRMB-TADAH	1.66	6.7	3.51	96.8	89.7
+C2f-iRD-C2f-iRMB-DySample-TADAH	1.68	6.7	3.54	97.3	90.0
+C2f-iRD-C2f-iRMB-DySample-TADAH-F–PIoUv2	1.68	6.7	3.54	97.9	90.8

3.2.5. Comparison of Detection Performance of Different Models

To further evaluate the effectiveness and generalization ability of the proposed YOLOv8n–TiD model, a comparative analysis was conducted against several mainstream object detection algorithms, including Faster R-CNN, EfficientDet, and a series of YOLO-based lightweight models on the same tomato ripeness detection test set. The results are presented in Table 8. To ensure a fair and interpretable comparison, all baseline models were trained and evaluated using the same dataset partition, input resolution, and evaluation metrics. Compared with other single-stage detectors, YOLOv8n–TiD achieved the highest mAP and demonstrated notable lightweight characteristics. Specifically, it outperformed the original YOLOv8n model by a 1.9 percentage point increase in mAP, while reducing parameter count by 45%, FLOPs by 19%, and model size by 41%. Among the YOLO variants, YOLOv11n exhibited slightly lower parameter count and FLOPs compared to YOLOv8n–TiD. However, its model size was larger, and detection accuracy was inferior, indicating that YOLOv8n–TiD strikes a better balance between detection performance and model compactness. In contrast, two-stage detectors such as Faster R-CNN and EfficientDet did not perform well on the custom tomato dataset. Their relatively higher complexity and poor adaptability to dense, occluded, and fine-grained agricultural targets resulted in lower mAP values and inefficient resource usage, further highlighting the advantages of YOLO-based detectors in such agricultural vision tasks. These results demonstrate that YOLOv8n–TiD is a more suitable and efficient solution for real-time tomato ripeness detection in complex environments, offering both superior detection accuracy and a deployment-friendly model footprint.

Table 8. Experimental results of different model detections.

Models	Parameter/M	GFLOPs	Model Size/MB	mAP$_{@0.5}$ (%)	mAP@0.5:0.95 (%)	Detect Times/ms
EfficientDet	27.3	58.2	15.0	87.2	72.3	29.5
Faster R-CNN	137.75	403.8	107.8	62.3	53.8	70.5
YOLOv5s	2.36	6.5	4.72	94.2	78.3	40.3
YOLOv7	36.51	103.4	71.5	93.1	79.1	42.1
YOLOv8n	3.01	8.2	5.98	96.0	88.7	25.6
YOLOv10n	2.37	6.5	5.50	95.1	86.3	24.7
YOLO11n	1.56	4.5	5.23	93.2	87.4	20.1
YOLOv8n–TiD	1.68	6.7	3.54	97.9	90.8	19.4

To further evaluate the performance of the proposed YOLOv8n–TiD model under complex environmental conditions such as fruit overlap, foliage occlusion, and small object detection, a series of comparative experiments were conducted against several mainstream detectors, including EfficientDet, Faster R-CNN, and multiple variants of the YOLO family. Figure 12 illustrates the visual detection outcomes between YOLOv8n–TiD and its related variants (YOLOv8n, YOLOv10n, and YOLOv11n), while Table 8 summarizes the quantitative metrics. The comparison reveals that as a single-stage detector, EfficientDet presents a significantly higher parameter count and computational complexity compared to the YOLO series. Its backbone and BiFPN structure exhibit limited capability in extracting and fusing high-frequency features, leading to poor discrimination between immature tomatoes and background foliage, and ultimately unsatisfactory detection performance, particularly for underdeveloped fruits. Furthermore, its inference speed lags behind that of recent lightweight YOLO models. Faster R-CNN, though equipped with a deep backbone, suffers from severe feature map resolution degradation due to multiple downsampling operations. As a result, boundary features are blurred or merged, causing missed detections and redundant bounding boxes. Notably, it demonstrated the slowest inference speed among all compared models. YOLOv5s exhibited frequent false detections, especially when fruits were intertwined with branches or leaves. This is primarily attributed to ambiguous positive-negative sample boundaries during training, which hindered the model's discrimination capacity. While YOLOv7 integrated more advanced feature fusion mechanisms, it failed to distinguish subtle differences between closely situated fruits, leading to bounding box misalignment. Both YOLOv5s and YOLOv7 showed no clear advantage in terms of detection speed. YOLOv8n showed degraded feature representation in complex environments. It struggled with detecting distant small targets and occluded fruits, largely due to its fusion module focusing excessively on global semantics while lacking fine-grained texture response. Consequently, features of adjacent fruits with similar ripeness levels were overly compacted, resulting in misidentification or inaccurate localization. YOLOv10n, which introduced dynamic attention mechanisms and multi-scale fusion, improved attention toward target regions to some extent. However, in densely packed and high-similarity environments, the model was prone to being distracted by pseudo-cues such as highlights or leaf vein textures, which weakened the model's ability to emphasize subtle maturity differences. This led to bounding box merging or drift, especially under occlusion and stacking scenarios. YOLOv11n, due to aggressive downsampling and channel compression, suffered from discriminative feature loss at P3–P5 scales, resulting in poor small object preservation. Furthermore, it lacked deformable alignment and robust attention mechanisms, making it susceptible to interference from background structures like veins and highlights. The sample reweighting and NMS stages further aggravated misdetections and false suppression, particularly in overlapping fruit scenarios. In contrast, the proposed YOLOv8n–TiD model demonstrated superior performance in detecting distant small fruits, distinguishing closely located tomatoes with similar ripeness, and effectively handling occlusion and stacking. Benefiting from multiple lightweight structural enhancements, YOLOv8n–TiD achieved a favorable trade-off between accuracy and complexity, showing the lowest model complexity and best detection performance among all compared models. These features make it especially suitable for deployment on mobile or resource-constrained platforms, affirming its practical value in intelligent agricultural applications.

Figure 12. Visual comparison of detection results obtained by different models under representative challenging orchard scenarios. The red area represents the detection advantage of YOLOv8n–TiD model The examples include: (**a**) distant small tomato targets; (**b**) fruit stacking and overlapping; (**c**) adjacent ripeness stages with similar visual appearance; and (**d**) branch and leaf occlusion.

3.2.6. Analysis of Model Generalization Ability

To comprehensively evaluate the detection capability and cross-variety generalization of the model, an additional tomato type—cherry tomato—was used for external validation. During image acquisition, the diversity of the dataset was ensured by including as many tomato fruits as possible in each image while capturing complex orchard conditions such as strong illumination, backlighting, dense fruit clusters, and occlusion by leaves and branches. These settings increase dataset complexity and detection difficulty. Notably, the maturity stages of this cherry tomato variety fully correspond to the four-level maturity grading scheme adopted in this study. Compared with large tomatoes, cherry tomatoes feature smaller fruit size and denser growth patterns, with numerous fruits distributed on a single truss exhibiting different maturity levels simultaneously. Moreover, the fruit clusters do not naturally hang freely; instead, they are frequently intertwined with branches and leaves, with clustered fruits often overlapping. Such characteristics impose greater challenges for detection, making this tomato variety suitable for evaluating the generalization performance of the proposed model.

As shown in Figure 13, the detection performance of YOLOv8n and YOLOv8n–TiD was compared. The results indicate that the baseline model exhibits insufficient detection capability. Due to inter-varietal differences in color distribution, surface gloss, and micro-texture, domain shift occurs, and the YOLOv8n backbone struggles to preserve fine-grained chromatic and textural cues at deeper downsampling scales (P3–P5), leading to difficulty in distinguishing immature tomatoes from green leaves. In addition, occlusion disrupts shape and color continuity, amplifying misalignment between classification and localization in decoupled heads, while attention mechanisms are prone to being misled by leaf venation or reflective artifacts. Meanwhile, insufficient representation of the new fruit type in training data and data-augmentation bias against small targets reduce the effective sampling weight of occluded instances; during inference, suppression of densely overlapped proposals exacerbates missed detections and false removals.

Figure 13. Visual comparison of YOLOv8n and YOLOv8n–TiD on the external cherry tomato validation dataset. The red area represents the detection advantage of YOLOv8n–TiD model The figure shows: (**a**) the original cherry tomato image; (**b**) detection results of YOLOv8n; and (**c**) detection results of YOLOv8n–TiD under cross-variety ripeness detection conditions.

In contrast, YOLOv8n–TiD demonstrates markedly superior performance. The newly designed feature extraction module achieves a balance between large receptive fields and high-fidelity low-level details, reducing texture and edge loss for small fruits under deep subsampling and enhancing fine-grained cues such as color, gloss, and micro-texture. This promotes greater inter-class separation and prevents mutual suppression among adjacent fruits with similar maturity levels. The TADAH detection head enforces consistency between class prediction and bounding localization in occluded and densely clustered scenes, effectively mitigating duplicate boxes and localization bias. DySample alleviates aliasing and semantic drift during upsampling, reinforcing the reconstruction of high-frequency structures such as stems and leaf edges and improving contour completion and instance separation under occlusion. The F–PIoUv2 loss applies focal re-weighting to small, occluded, and visually similar targets and enhances quality-consistent optimization, improving recall of subtle true positives and reducing erroneous suppression among highly overlapping proposals.

Overall, the synergistic incorporation of structural inductive priors and hard-sample-oriented optimization significantly enhances robustness and transferability across tomato varieties, achieving higher detection accuracy and more stable localization consistency under composite challenging scenarios, while maintaining low model complexity and computational cost.

3.2.7. Comparison of Model Performance Before and After Improvement

Figure 14 presents the evaluation results of the original and improved models using two levels of mAP. For mAP@0.5, YOLOv8n–TiD consistently outperforms the baseline starting from the 100th training epoch and maintains its superiority until the end of training. For mAP@0.5:0.95, the improved model surpasses the original model from around the 150th epoch onward and continues to lead thereafter. Overall, the detection accuracy of YOLOv8n–TiD remains at a high level throughout training, even under a lightweight design, demonstrating the effectiveness and robustness of the proposed improvements on the custom tomato ripeness dataset.

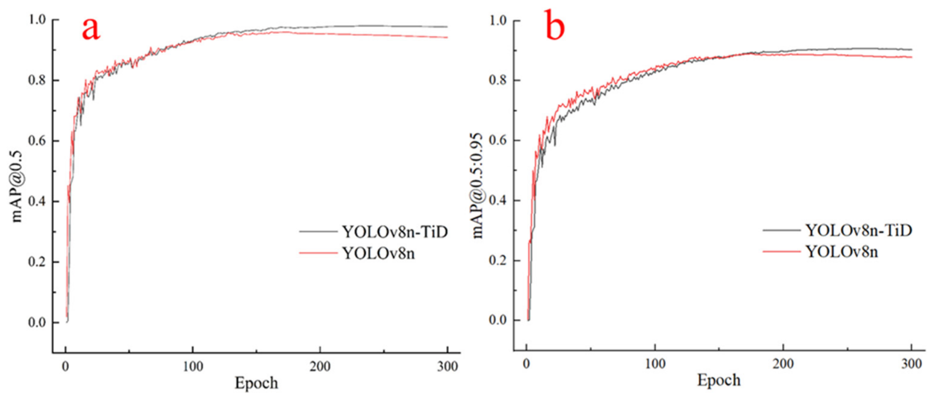

Figure 14. Comparison of validation mAP curves between YOLOv8n and YOLOv8n–TiD during training: (**a**) mAP@0.5; and (**b**) mAP@0.5:0.95.

As shown in Figure 15, both the training and validation losses decrease monotonically with increasing iterations and gradually stabilize in the later stages, indicating a well-converged optimization process. This trend demonstrates that the model effectively fits the training data without exhibiting any rebound in validation loss, suggesting that no obvious overfitting or underfitting occurred during training. The smooth convergence of the loss curves further reflects the stability of the training process and the appropriateness of the parameter adjustments, contributing to the overall robustness and reliability of the model.

As shown in Table 9, YOLOv8n–TiD achieved more balanced detection performance across the four tomato ripeness categories than the original YOLOv8n. From the perspective of class-wise precision, YOLOv8n–TiD improved all four categories, with the most pronounced gains observed for immature and breaker tomatoes. This indicates that the improved model can better suppress false detections caused by leaf-like backgrounds, occlusion, and visually similar transitional color features. In terms of recall and F1-score, YOLOv8n–TiD also showed clear advantages for immature, breaker, and turning categories, increasing the F1-score from 87.1% to 95.9%, 90.3% to 96.8%, and 91.1% to 95.2%, respectively. These results suggest that the proposed modules enhance the model's ability to distinguish adjacent ripeness stages and detect partially occluded or densely distributed fruits.

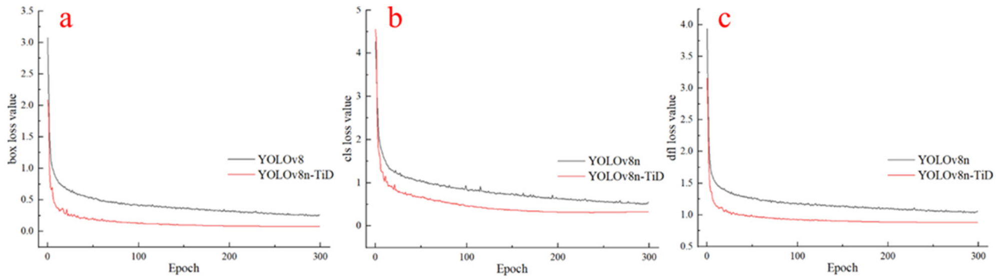

Figure 15. Training loss curves of YOLOv8n–TiD on the tomato ripeness dataset: (**a**) box loss for bounding-box regression; (**b**) classification loss for ripeness category prediction; and (**c**) DFL loss for bounding-box distribution learning.

Table 9. Comparison of detection across different maturity categories.

Models	Class	Precision/%	Recall/%	F1(%)	$mAP_{@0.5}$ (%)	mAP@0.5:0.95 (%)
YOLOv8n	immature	86.8	87.5	87.1	96.1	86.2
	breaker	85.3	95.9	90.3	96.7	89.7
	turning	88.5	93.8	91.1	96.2	91.4
	mature	90.8	96.2	93.4	91.2	88.5
YOLOv8n–TiD	immature	94.3	97.6	95.9	98.9	87.2
	breaker	96.5	97.2	96.8	98.8	91.8
	turning	93.8	96.6	95.2	98.2	91.5
	mature	92.5	87.6	90.0	94.8	89.5

For the mature category, YOLOv8n–TiD achieved higher precision and higher AP values, but its recall decreased compared with YOLOv8n. This indicates that the improved model produced fewer false-positive detections for mature tomatoes but missed a small number of mature instances. This phenomenon may be related to the relatively limited number of mature samples and the visual variability caused by illumination, occlusion, and fruit overlap. Nevertheless, YOLOv8n–TiD still increased the AP@0.5 of the mature category from 91.2% to 94.8% and AP@0.5:0.95 from 88.5% to 89.5%, demonstrating improved localization quality and ranking performance.

Overall, the average F1-score of YOLOv8n–TiD across the four ripeness categories reached 94.5%, which was 4.0 percentage points higher than that of YOLOv8n. The mean AP@0.5 and mean AP@0.5:0.95 were also improved from 95.1% to 97.7% and from 89.0% to 90.0%, respectively. These class-wise results further confirm that the performance improvement of YOLOv8n–TiD is not only reflected in global metrics but also in the enhanced recognition stability of visually similar and imbalanced ripeness categories.

Figure 16 presents the normalized confusion matrices of YOLOv8n–TiD and YOLOv8n, providing an intuitive category-level comparison of their recognition performance across different tomato ripeness stages. Overall, YOLOv8n–TiD achieves higher diagonal values than YOLOv8n for the four ripeness categories, namely mature, turning, breaker, and immature, indicating improved correct classification capability. Specifically, the correct recognition proportions of mature, turning, breaker, and immature tomatoes increase from 0.83, 0.94, 0.92, and 0.95 for YOLOv8n to 0.88, 0.97, 0.94, and 0.98 for YOLOv8n–TiD, respectively. These results suggest that the proposed model enhances the discriminative feature representation of tomato fruits at different ripeness stages, especially under complex conditions involving similar fruit colors, foliage occlusion, and dense fruit distribution.

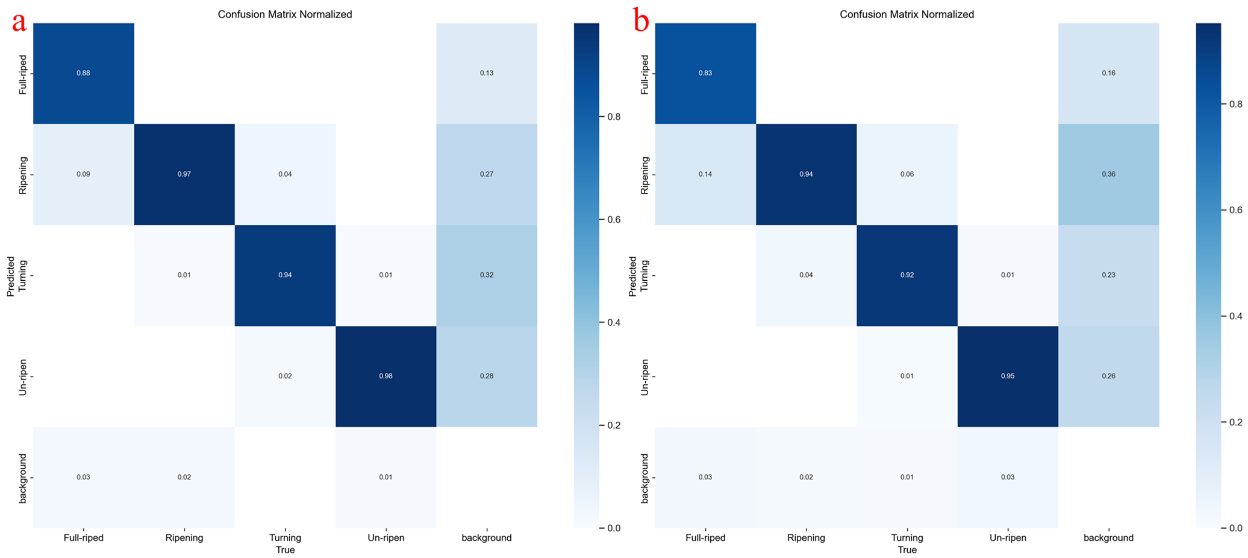

Figure 16. Normalized confusion matrices of the two models on the tomato ripeness test set: (**a**) YOLOv8n–TiD; and (**b**) YOLOv8n.

For inter-class confusion between adjacent ripeness stages, YOLOv8n–TiD also demonstrates stronger suppression capability. For example, the proportion of mature tomatoes misclassified as turning decreases from 0.14 in YOLOv8n to 0.09 in YOLOv8n–TiD, indicating that the improved model can more effectively distinguish mature tomatoes from turning-stage tomatoes. Similarly, the confusion between turning and breaker is also reduced: the proportion of turning tomatoes misclassified as breaker decreases from 0.04 to 0.01, while the proportion of breaker tomatoes misclassified as turning decreases from 0.06 to 0.04. This indicates that the improved feature extraction structure and task-adaptive detection head enhance the model's perception of subtle color gradients and surface texture differences during ripeness transition, thereby reducing misclassification between adjacent maturity stages.

In addition, YOLOv8n–TiD shows an advantage in reducing missed detections. The proportion of immature tomatoes misclassified as background decreases from 0.03 in YOLOv8n to 0.01 in YOLOv8n–TiD, and the misclassification of breaker tomatoes as background is also reduced. This suggests that the improved model can better preserve fine-grained features of small or partially occluded targets and reduce the probability of incorrectly treating tomato instances as background. Although some background regions are still incorrectly activated as tomato categories, the overall confusion matrix demonstrates that YOLOv8n–TiD produces more concentrated diagonal responses and lower inter-class confusion than YOLOv8n. These results further indicate that the advantage of YOLOv8n–TiD is reflected not only in global detection metrics but also in improved recognition stability for visually similar and imbalanced tomato ripeness categories.

To evaluate the cross-variety transferability of the proposed model, an external cherry tomato dataset was constructed for independent validation. This dataset contained 753 images and 6369 annotated tomato instances, including 2434 mature, 1195 turning, 820 breaker, and 1920 immature tomatoes. Compared with the large-fruit tomato dataset used for model training, cherry tomatoes are generally smaller, more densely distributed within fruit clusters, and more frequently affected by fruit overlap, branch interference, and leaf occlusion. These characteristics increase the difficulty of ripeness detection and provide a suitable external dataset for assessing the model's adaptability under cross-variety conditions.

YOLOv8n and YOLOv8n–TiD were evaluated on the external cherry tomato dataset under the same testing protocol. As shown in Table 10, both models exhibited lower detection performance than that obtained on the original large-fruit tomato dataset, indicating the presence of a domain shift between different tomato varieties. Nevertheless, YOLOv8n–TiD achieved improved recall for the immature, turning, and mature categories, increasing from 84.1%, 74.2%, and 88.1% to 89.1%, 78.3%, and 88.3%, respectively. This suggests that the proposed model can retain more true-positive detections for small and densely distributed cherry tomato targets. In particular, the immature category showed clear improvements in precision, recall, and F1-score, increasing from 80.3%, 84.1%, and 82.2% to 82.9%, 89.1%, and 85.9%, respectively. This result indicates that the improved feature extraction and dynamic feature fusion mechanisms enhance the model's ability to distinguish green immature fruits from leaf-like backgrounds under cross-variety conditions.

Table 10. Cross-variety validation results of YOLOv8n and YOLOv8n–TiD on the external cherry tomato dataset.

Models	**Class**	**Precision/%**	**Recall/%**	**F1(%)**	**mAP$_{@0.5}$ (%)**	**mAP$_{@0.5:0.95}$ (%)**
YOLOv8n	immature	80.3	84.1	82.2	90.8	69.0
	breaker	71.1	79.9	75.2	78.9	65.0
	turning	74.1	74.2	74.1	82.7	67.2
	mature	90.0	88.1	89.0	93.0	68.9
YOLOv8n–TiD	immature	87.6	89.1	88.3	91.3	77.1
	breaker	70.5	73.8	72.1	78.8	66.9
	turning	71.9	78.3	75.0	79.9	67.0
	mature	87.6	88.3	87.9	93.2	70.7

From the perspective of localization quality, YOLOv8n–TiD achieved higher AP@0.5:0.95 in most categories. Specifically, AP@0.5:0.95 increased from 69.0% to 77.1% for immature tomatoes, from 65.0% to 66.9% for breaker tomatoes, and from 68.9% to 70.7% for mature tomatoes. The average AP@0.5:0.95 across the four categories increased from 67.5% to 70.4%, indicating that YOLOv8n–TiD provides more stable bounding-box localization under stricter IoU thresholds. This improvement may be attributed to the task-adaptive detection head and F–PIoUv2 loss, which enhance spatial alignment and regression optimization for small, occluded, and densely overlapped fruits.

It should also be noted that YOLOv8n–TiD did not outperform YOLOv8n in all category-level metrics. For example, the AP@0.5 values of the breaker, turning, and mature categories slightly decreased compared with YOLOv8n, suggesting that cross-variety differences in fruit size, color distribution, surface texture, and cluster morphology still affect model transferability. Therefore, the external validation results should be interpreted as preliminary evidence of cross-variety adaptability rather than definitive proof of complete generalization. Overall, YOLOv8n–TiD demonstrates improved recall and localization quality on the cherry tomato dataset while maintaining a lightweight structure; however, further validation using larger multi-variety datasets is still needed to strengthen the evidence for its generalization capability.

To intuitively evaluate the improvement effect of YOLOv8n–TiD over YOLOv8n, the detection results from the penultimate layer of the network are visualized as heatmaps, as shown in Figure 17. These heatmaps can be considered representations of the model's perceptual attention. When performing detection, YOLOv8n exhibits weak feature perception and a limited receptive range for tomato fruits, with only sparse attention focused on the fruits. Moreover, it often misidentifies leaves as unripe tomatoes, leading to incomplete detection, poor recognition accuracy, and inaccurate bounding box localization. In contrast, YOLOv8n–TiD demonstrates significantly improved performance, with most attention focused precisely on the tomato fruits. This is due to the introduction of the C2f-iRD module, which leverages re-parameterized multi-dilation convolution and inverted residual units to notably expand the effective receptive field and enhance the fidelity of low-level textures and edges, thereby preserving discriminative details for small tomatoes at shallow levels such as P3. Additionally, the lightweight C2f-iRMB neck enables efficient cross-scale feature fusion, enhancing the prominence of subtle color variations and surface gloss, which helps widen the feature gap between neighboring fruits of similar ripeness and suppresses their mutual merging. The TADAH detection head, based on task-adaptive dynamic alignment, allows decoupled classification and regression branches to adaptively align with their respective difficulty levels, mitigating the misalignment between localization and classification caused by occlusion and clustering and improving boundary consistency and multi-object separability. During upsampling, DySample reduces aliasing and semantic drift while reinforcing high-frequency detail reconstruction such as stems, contours, and leaf edges, facilitating the completion of occluded targets and the separation of overlapping fruit boundaries. Meanwhile, F–PIoUv2 implements focalized quality regression and IoU distribution optimization, applying higher gradient weights to hard samples and small targets, thereby improving recall of low-score true positives and reducing the suppression of highly overlapping candidates during post-processing. Collectively, these mechanisms synergistically enhance the model's ability to maintain high detection accuracy and robust positional consistency under complex scenarios involving distant small targets, adjacent fruits with similar ripeness, fruit clustering, and foliage occlusion—all while maintaining a lightweight design with low parameter and computational costs.

Figure 17. Heatmap comparison between YOLOv8n and YOLOv8n–TiD for tomato ripeness detection. Cloud map represents the detectable range of the model. The figure shows: (**a**) original image; (**b**) heatmap generated by YOLOv8n; and (**c**) heatmap generated by YOLOv8n–TiD.

3.2.8. Edge Device Deployment

To evaluate the feasibility of deploying YOLOv8n–TiD on mobile edge devices, the trained model was deployed on an Android smartphone, and a real-time tomato ripeness detection application was developed. The test device was a Xiaomi 13 equipped with a Snapdragon 8 Gen 2 processor, 12 GB RAM, and the Android 14 operating system. Model deployment was implemented using the NCNN inference framework, which is optimized for mobile neural network inference and enables efficient model execution under limited computational resources. In this study, the NCNN model was configured for FP32 inference, and Vulkan acceleration was enabled to improve mobile-side inference efficiency. Specifically, the trained PyTorch model weights were first exported to ONNX format and then converted into the param and bin files required by the NCNN toolchain. During model conversion, operator compatibility adjustment and model simplification were performed. The converted model files were then integrated into an Android application using Android Studio, enabling real-time mobile inference and visualization of detection results.

In the mobile-side performance test, the input resolution was set to 640 $\times$ 640. The results showed that the deployed YOLOv8n–TiD model achieved a model-side average inference latency of 28.6 ms on the Xiaomi 13 (Xiaomi Corporation) device, with a p95 latency of 37.2 ms. Meanwhile, the real-time application maintained an average display frame rate of approximately 30 FPS. These results indicate that the model can maintain stable real-time inference performance on the tested mobile device and basically meet the real-time requirements of tomato ripeness detection in natural scenes. The relatively low average inference latency and acceptable p95 latency further suggest that no obvious inference stuttering occurred during continuous video-stream detection, demonstrating the potential of the model for edge-device deployment.

To further observe the practical detection performance on the mobile device, tomato ripeness detection was conducted under both real-time video-stream conditions in natural environments and static-image conditions, as shown in Figure 18. Overall, YOLOv8n–TiD was able to identify and localize tomatoes at different ripeness stages on the Android device, and the detection process was relatively smooth. However, compared with static-image detection, the confidence scores obtained from real-time video streams were slightly lower. This was mainly related to frame-level quality fluctuations in mobile inputs. Static images usually have higher resolution, more uniform illumination, and clearer backgrounds, whereas real-time video frames are more susceptible to illumination changes, motion blur, camera shake, automatic exposure adjustment, and compression noise. These factors reduce the clarity and feature discriminability of tomato target regions, causing certain shifts in the feature representations extracted by the model and thereby affecting the output confidence scores.

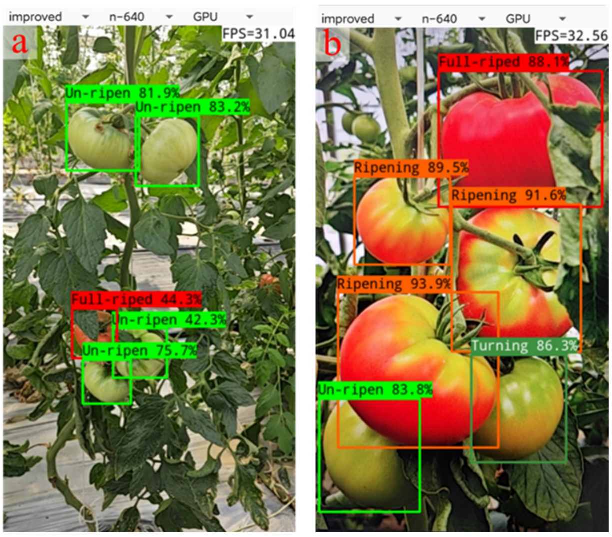

Figure 18. Mobile-device detection results of YOLOv8n–TiD on an Android smartphone: (**a**) real-time video-stream detection; and (**b**) static-image detection.

4. Discussion

This research addresses the critical need for efficient tomato ripeness assessment in natural agricultural settings through the development of YOLOv8n–TiD—an optimized detection framework building upon YOLOv8n. Our architectural innovations include the integration of re-parameterized dilated convolutions and inverted residual blocks within the C2f structure, substantially augmenting the effective receptive field while preserving computational efficiency. The synergistic combination of DySample upsampling with the TADAH detection head establishes coordinated feature refinement across both spatial and task domains, effectively countering misalignment and localization inaccuracies. Furthermore, the F–PIoUv2 loss function introduces interval mapping and adaptive penalty mechanisms that enhance learning dynamics and convergence behavior for intermediate-quality predictions.

Empirical evaluations demonstrate compelling performance gains: our model achieves a 45% reduction in parameterization, a 19% decrease in computational load, and 41% compression in model storage, while simultaneously elevating mean average precision by 1.9 percentage points. Cross-varietal validation using a distinct cherry tomato cultivar confirms the framework's generalization capacity. Comprehensive analysis across diverse challenging scenarios underscores the method's robustness. Practical deployment on Android platforms via the NCNN inference engine sustains real-time throughput exceeding 30 FPS, fulfilling operational requirements for resource-constrained edge computing environments.

By systematically optimizing both architectural compactness and deployment readiness, this work establishes a viable pathway for implementing high-performance vision systems in precision agriculture. The proposed methodology contributes meaningful advances toward practical object detection solutions while providing valuable design principles for intelligent agricultural technology development.

5. Conclusions

Compared with two-stage object detection frameworks and efficient detectors such as EfficientDet, the proposed model achieves a better trade-off among detection accuracy, model complexity, and deployment feasibility. Notably, in complex agricultural scenarios—such as dense fruit clusters, visually similar ripeness stages, and severe occlusions by foliage—our model demonstrates superior recall and more stable bounding box localization. However, several limitations remain. Firstly, the dataset used for training is primarily composed of images captured from a specific tomato cultivar under fixed acquisition conditions, which restricts the model's generalization capability across different cultivars with varying appearance traits. Secondly, due to the inherent capacity limitations of lightweight models, the ability to capture fine-grained features in ultra-high-resolution frames is constrained. Lastly, heterogeneous hardware platforms often have inconsistent operator acceleration support, which may hinder the theoretical inference speed gains promised by quantized models. In future work, we plan to explore advanced model compression techniques such as pruning and knowledge distillation to further reduce model parameters and computational overhead while maintaining detection accuracy, thus enabling efficient deployment on mid- to low-end edge devices. Additionally, we aim to incorporate multi-modal sensor fusion to develop a comprehensive quality assessment system for tomatoes. Building upon the ripeness detection foundation, we will also investigate the correlation between tomato phenotypes and internal quality attributes, with the goal of constructing a unified algorithm for ripeness and quality evaluation.

Author Contributions: Writing—review & editing, Writing—original draft, Methodology, Investigation, Conceptualization, Formal analysis, J.D.; Investigation, Visualization, Validation, Y.Z.; Investigation, Data curation, Y.W.; Investigation, Data curation, L.H.; Writing—review & editing, Y.X.; Resources, R.Z.; Project administration, Funding acquisition, X.X. All authors have read and agreed to the published version of the manuscript.

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Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

YOLO	You Only Look Once
C2f	Cross-Stage Partial with two feature fusion branches
iRMB	Inverted Residual Mobile Block
DRB	Dilated reparam block
TADAH	Task-adaptive dynamic alignment head
F–PIoUv2	Focaler Powerful-IoU v2
DCNv2	Deformable Convolutional Networks v2
IoU	Intersection over Union
FPS	Frames Per Second

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